

Usability Testing: ByteBreakers

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Objectives

Ease of Use and Workflow Efficiency: Look at how the tasks flow; things like sign-ups and posting bounties should feel quick and not clunky. See how naturally people move through tasks like creating an account, setting up 2FA, or navigating the dashboard.

Security Perception: Pay attention to how the security elements come across. Do they give a sense of safety or feel like extra hurdles?

Pain Points and User Feedback: Pick up on how people talk about clarity, whether things are easy to follow, and how secure the whole experience feels. And get a rating on a scale of 1-10. Notice where people pause, fumble, or look confused, those small stumbles often say more than they realize.

UI/UX Comfort and Actionable Recommendations: Think about whether the design feels natural to interact with or if anything gets in the way of just using it. Keep track of what users say and what they actually do, usually those little moments that point to what really needs fixing.

Methodology:

The study involves 6–8 participants with a balanced mix of technical and non-technical backgrounds. Sessions will include audio and screen recording (with consent) to capture them. We conducted a pre-interview survey form, followed by qualitative interviews and analysis.

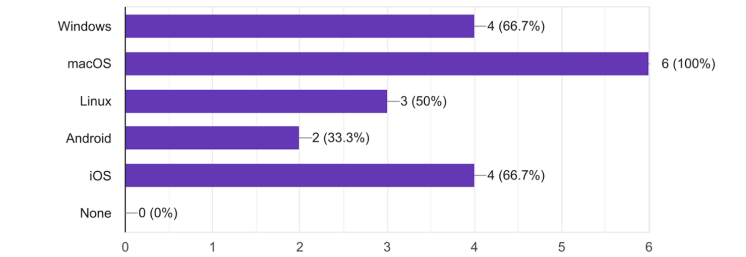
Data Collection & Analysis The evaluation combined task timing, error tracking, and success rates with user comments and impressions on security, layout, and usability. Screen and audio recordings helped uncover patterns, with findings summarized into clear, practical insights. Data was then transcribed, and qualitative coding and analysis were performed.

Participant Demographics

Participants aged 23-26 generally expressed high digital privacy concerns, with one exception indicating low concern. Most participants were enrolled in MS and one in a PhD. Those who were *Very* or *Extremely* familiar with web applications tended to report higher privacy concern levels, suggesting a link between awareness and caution. While device count didn't consistently predict concern, users with 2-3 devices showed slightly elevated caution. Browser diversity showed mixed patterns, with both high and low concerns among users with three or more browsers, indicating no direct correlation. However, participants using multiple operating systems did appear more wary, likely because their broader technical exposure made them more privacy-aware.

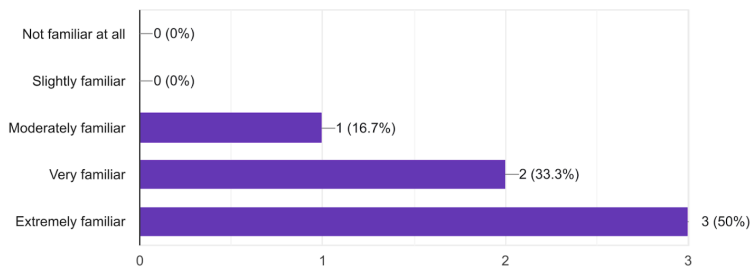
What operating system are you using?

6 responses



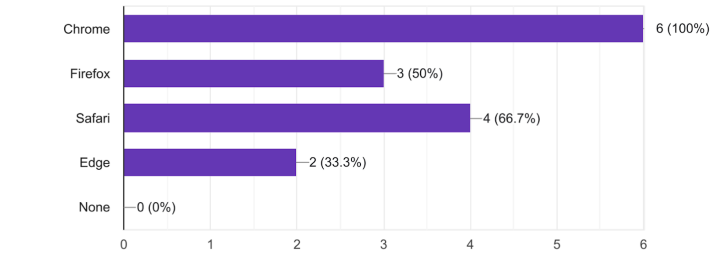
How familiar are you with web applications?

6 responses



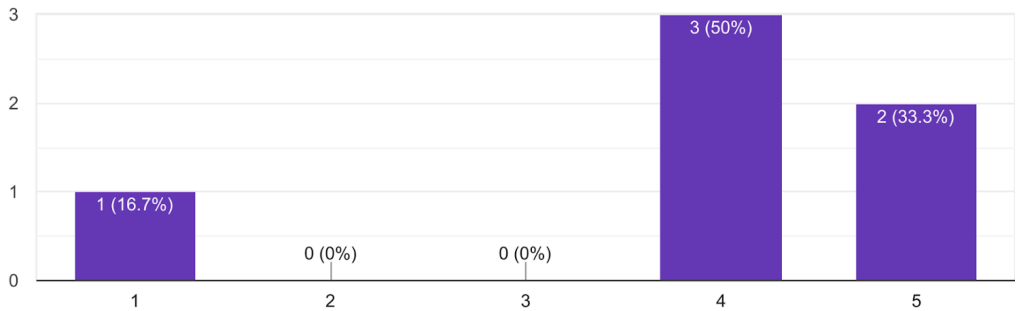
What browsers do you use?

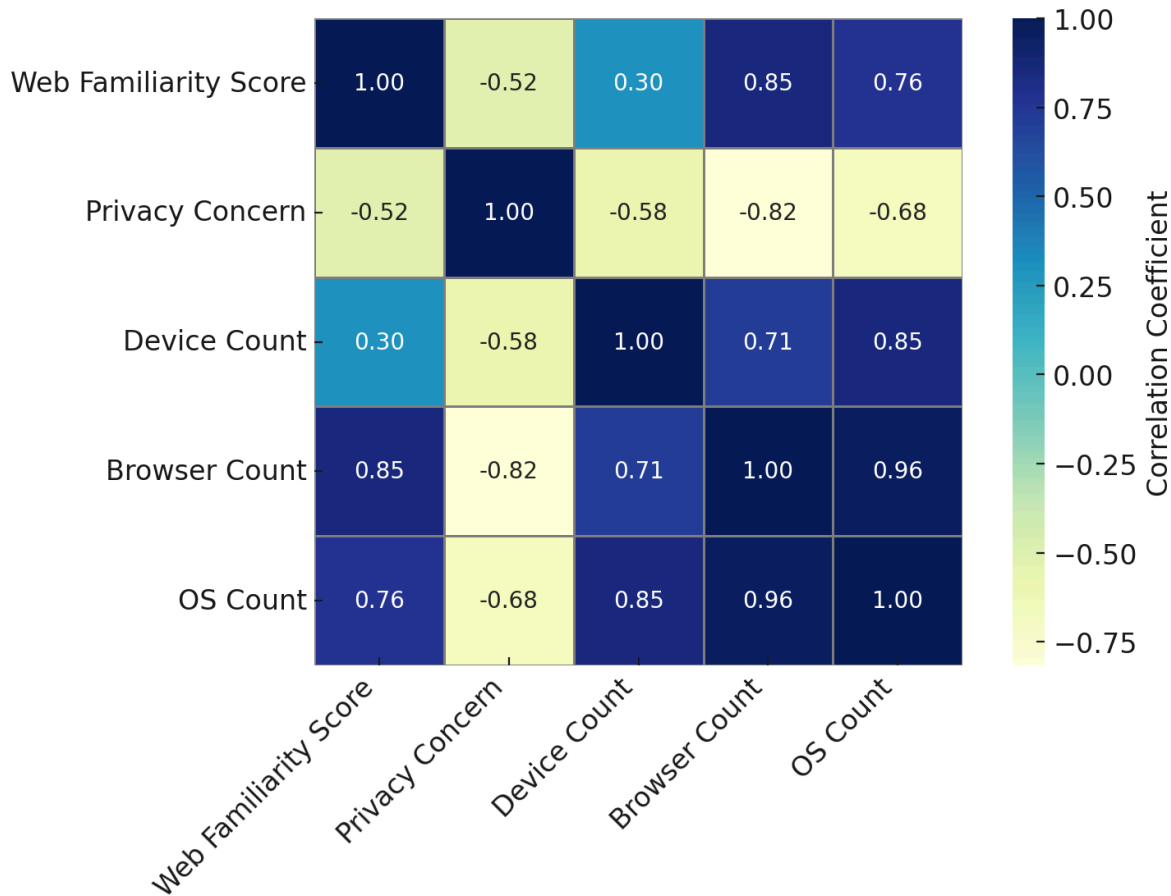
6 responses



On a scale of 1-5, how concerned are you about your digital privacy and security?

6 responses





Correlation Matrix

Clustering and Correlation :

Looking deeper, a few interesting connections emerged. People with more devices typically used more browsers, which makes intuitive sense, but what stood out more was the link between familiarity with web tools and concern about digital privacy; those who knew more worried more. Cybersecurity experience also seemed to matter: most of those with a background in the field described themselves as highly familiar with digital tools. However, not everyone with a technical background shared the same level of concern; one individual, in particular, appeared quite unconcerned, perhaps feeling confident in their own ability to manage digital risks.

Moving forward, two distinct clusters came up when we grouped users based on their digital habits and privacy attitudes. One group, tech-savvy and concerned, clearly leaned toward caution. While equally tech-comfortable, the other didn't show the same level of worry, hinting at differing personal thresholds for risk or trust in existing protections.

Pre-Test Briefing

Hello! Thanks so much for joining today's usability session for ByteBreakers. We really value your time and input. The goal is to see how people use the platform, what feels smooth and confusing, and where things can improve. Just to be clear, this isn't a test of your skills. We're simply looking at how the platform performs in a real-world setting.

Objectives

There are two main things we're hoping to learn from you today:

- How easy it is to move around the platform and complete tasks
- How people respond to built-in security features like two-factor authentication and secure fields
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Consent & Privacy

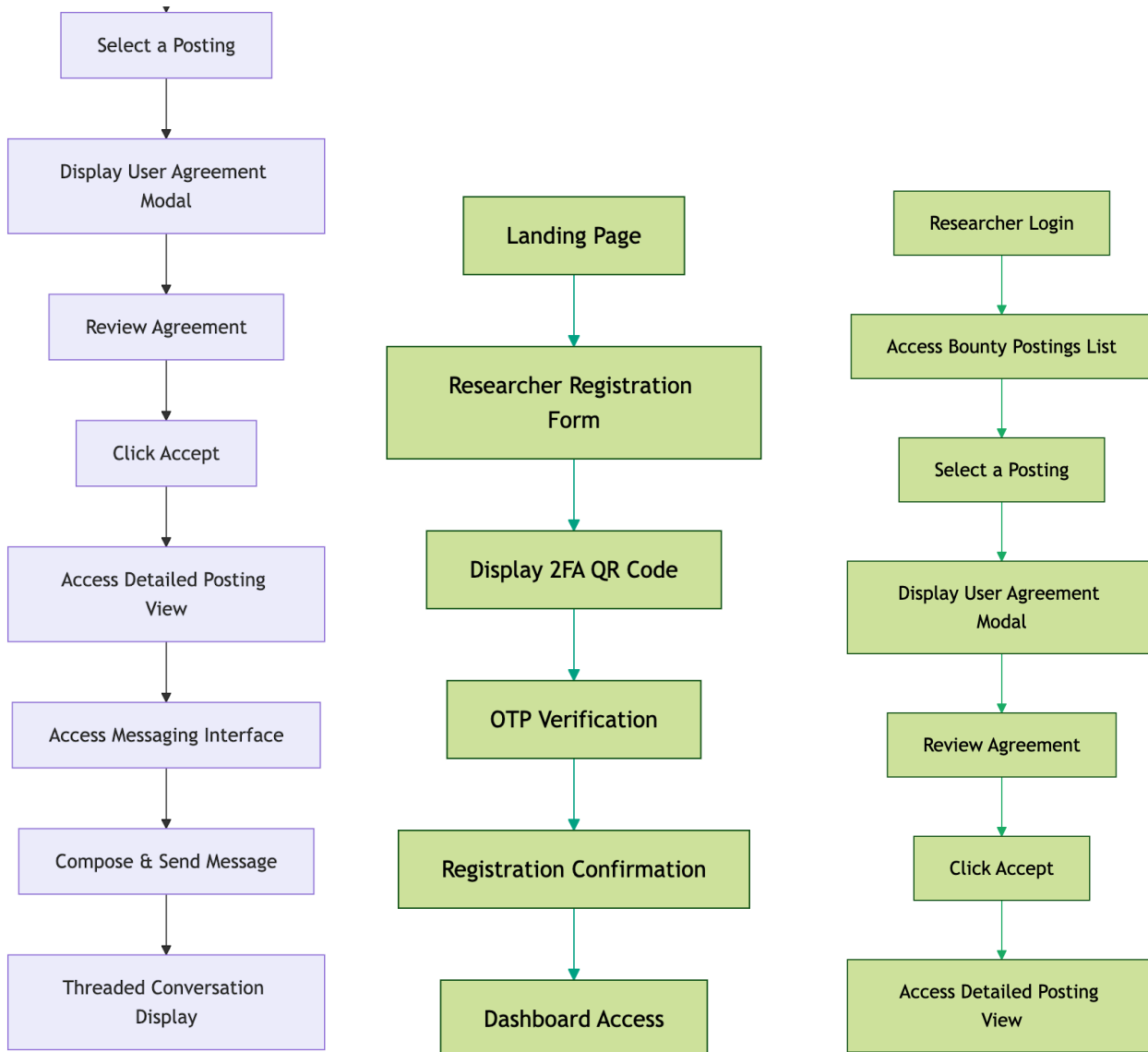
Before we begin, we'd like to confirm that you're okay with us recording your screen and audio during this session. This is just for internal review; it helps us understand user behavior more accurately. Everything you share will stay private and not be linked to your identity.

How the Session Will Go

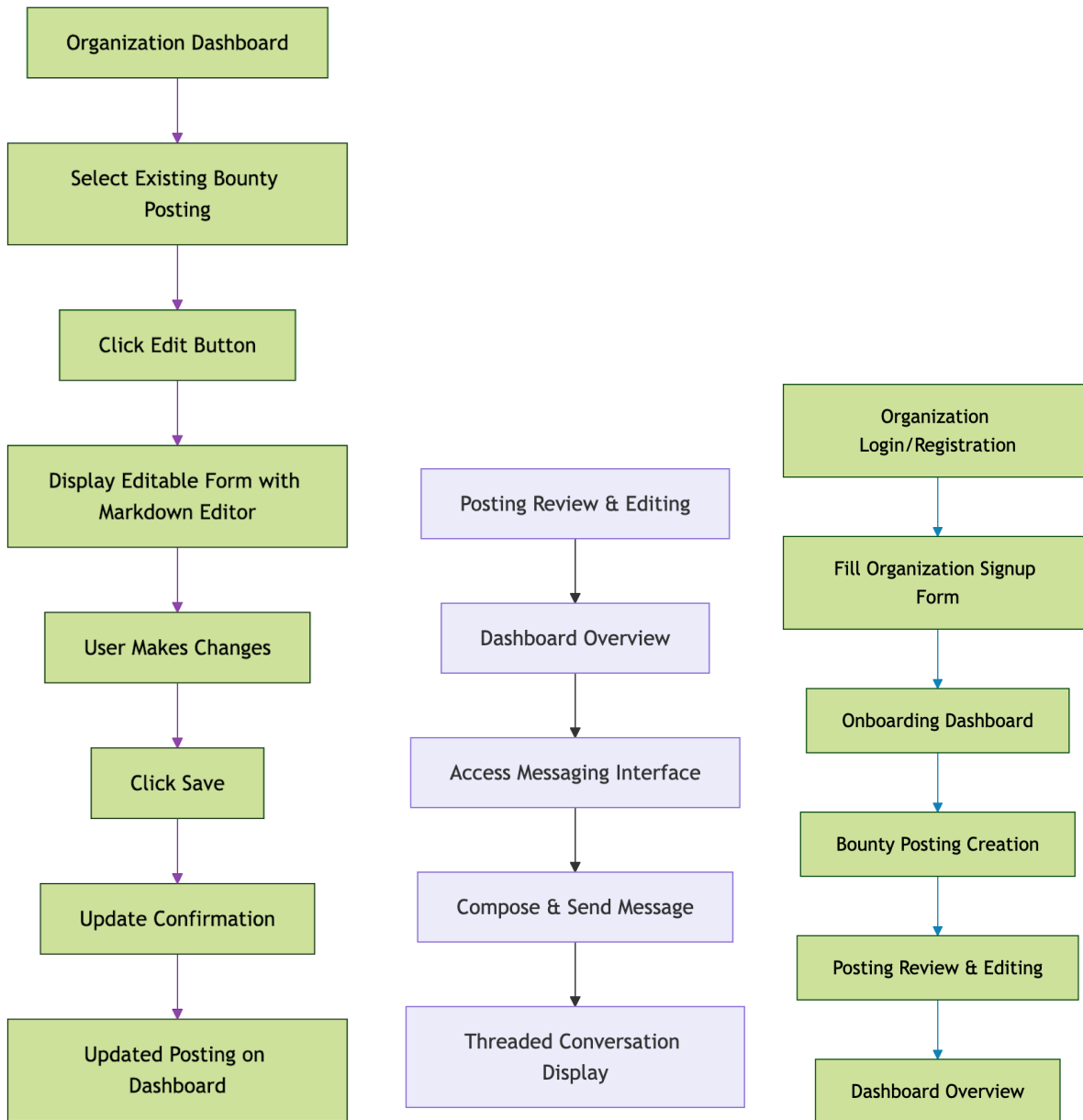
As you work through the tasks, feel free to talk out loud and share what you're thinking, what makes sense, or anything that throws you off. Don't worry about doing things the "right way"; your natural reactions are what we value most. We'll guide you through a series of tasks but won't provide hints or answers, and we're interested in seeing how you experience the platform naturally.

User Flow Charts:

Flow Diagrams for User Tasks:



Flow Diagrams for Organisation Tasks:



Test Scenarios & Tasks

Scenario 1: New User Registration (Researcher)

“Please register as a researcher. Enter your name, email, and password. When prompted, complete the 2FA setup by scanning the QR code with an authenticator app and entering the one-time code.”

This step looks at how users experience the sign-up flow, whether the form feels straightforward, if 2FA makes sense, and how secure or clunky it all feels.

Scenario 2: User Browsing & Agreement Acceptance

“Log in as a researcher, browse bounty postings, and accept the user agreement to view detailed information.” This scenario looks at whether users find the browsing interface easy to navigate and how they perceive the agreement prompt. It also checks how smoothly users move between the posting list and detail views.

Scenario 3: Organization Login with Researcher Credentials

“Attempt to log into the organization portal using researcher credentials.”

This one is simple but telling, it shows whether the system handles incorrect access attempts gracefully.

We’re looking for clarity in the feedback, whether users understand what’s going on, and how effective the role-based restrictions feel from the user side.

Scenario 4: Organization Registration & Onboarding

“Please register as an organization. Fill out the required fields, including your company name, email, and password. Proceed to the onboarding dashboard once registration is complete.” This scenario checks how smooth the registration feels from a user perspective. Security input handling and message clarity will also be considered.

Scenario 5: Creating and Managing Bounty Postings

“Log in as an organization, create and submit a bounty posting using the markdown editor, then confirm its visibility on the dashboard.” The focus is on how straightforward the editor feels, whether task instructions are understandable, and how users respond to any security prompts.

Scenario 6: Editing Existing Bounty Postings

“Locate and edit an existing bounty posting. Make changes and save the updated details using the markdown editor.” This task checks how easy it is to find the edit option, make updates, and understand the confirmation process. If users feel uncertain about the editing interface or the system response, it’s a red flag. Secure input handling is also part of the review.

Scenario 7: Secure Messaging

“Log in sequentially as researcher and organization. Send a secure message and reply to simulate a conversation.” This task evaluates whether the messaging interface feels intuitive and whether users believe their messages are private and encrypted. Observations will include layout usability, clarity of security indicators, and navigation across threads.

Scenario 8: New Organization Signup without Sample Data

“Sign up a new organization and immediately access the dashboard.” This scenario helps us see what users expect right after signup. If there’s no sample data shown, how does that affect their first impression? It’s also useful for spotting any confusion around messaging or onboarding flow.

Scenario 9: Archiving and Submission Workflow

“Log in as an organization, archive a bounty posting, edit it, and submit your changes.” This scenario tests how easy it is to archive content, make edits afterward, and receive system feedback. It also checks whether archived data is handled securely and updates are reflected correctly.

Scenario 10: Attempting Unarchive Operation

“Attempt to unarchive a previously archived posting.” This task gives insight into whether the system communicates limitations or restrictions clearly. If users try something that’s not allowed, the messaging should explain it well, not leave them guessing.

Scenario 11: Change Password for Organization and Relogin

“Navigate to account settings, change your organization’s password, then log out and log back in using the new password.” This step checks how easy it is to find the password change option, whether the process feels secure, and if the post-login behavior meets expectations. If something goes wrong, it’s important to see how well the platform communicates what to do next.

Post-Test Interview Questions

General Usability & Navigation

- If you had to give the platform a score between 1 and 10, where would you place it , and what made you choose that number?
- Would you feel comfortable recommending ByteBreakers to someone else, like a colleague or a friend? What influenced that decision?
- Were there any points during signup, posting, or messaging where you weren’t quite sure what to do next? Can you walk me through those moments?
- Was there any step or screen that felt more difficult or confusing than the others? What made it that way?
- Did the platform offer enough guidance while you were completing your tasks, or were you left guessing at any point?
- How did the navigation feel overall? Was it easy to move from one part of the platform to another, say, from signup to the dashboard or between bounty posts?
- Did any screen feel a bit too busy or require more effort than expected to get something done?

Security Perceptions

- How secure did you feel using the platform, especially in terms of your privacy and data?
- Did features like two-factor authentication or secure messaging help build trust, or did they feel like extra steps you weren’t sure you needed?
- Is there anything else you would want to see, like indicators or features, that would make you feel more confident about the platform’s security?
- Did the system clearly let you know when security features were being used , like during messaging or when uploading something?
- Were there any moments where you weren’t sure whether something was protected or not?

Error Handling & Feedback:

- When something didn't go as expected, say, during signup or posting , were the error messages helpful?
- Did the platform explain what the issue was and how to fix it, or did it leave you a bit confused?
- What would you change to make error messages or system feedback more useful?

UI/UX & Visual Design

- What was your impression of the overall design, things like layout, fonts, or color choices? Did it feel clean and modern, or a bit off?
- Were the buttons, links, and forms easy to spot and use?
- Did you notice whether the look and feel stayed consistent across different roles or screens?
- Did you feel that visual elements like icons or labels helped you understand actions or security features?

Detailed Functionality Feedback

- Were there any steps that felt unnecessary or confusing?
- Did the platform explain clearly what kind of information it needed and why?
- Did the system give you enough feedback after saving or submitting edits?
- How did the user agreement screen affect your experience , did it feel informative or more like a roadblock?
- Was the language in the agreement straightforward, or did anything feel overly technical or hard to digest?
- Anything else that you would like to mention?

TESTING RESULTS

Overall User Experience

Participants generally understood the platform's purpose ,a space for researchers and organizations to collaborate on bug bounty programs. While the concept came across clearly, the actual interaction experience didn't always live up to that clarity. Most users could navigate basic workflows, but several areas required more effort than expected. Some described the flow as intuitive in parts, yet disjointed or inconsistent in others.

Registration and Onboarding

User registration was mostly manageable, but some users faced issues with unclear prompts or system feedback. The onboarding flow worked better for researchers than for organizations. A few participants noted that once they landed on the organization dashboard, they weren't quite sure where to begin. There was a shared expectation for a more structured onboarding guide or at least subtle cues pointing toward first steps. For less technical users, this gap felt more noticeable.

Two-Factor Authentication Experience

The 2FA setup left mixed impressions. While users appreciated its presence from a security standpoint, several had difficulty understanding what to do after scanning the QR code. In some cases, it opened the password manager instead of prompting an OTP, which left users confused. Some suggested that clearer, inline instructions or a simple message explaining the next step would make a big difference.

Navigation and Interface Flow

Feedback pointed to a noticeable difference in polish between the researcher and organization dashboards. The researcher dashboard felt more user-friendly, while the organization interface was seen as static and less intuitive. Many wanted a sidebar or top menu to anchor their navigation. A few participants commented that without those visual anchors, it was easy to lose track of where you were or what actions were available.

Visual Design and Aesthetics

Most users described the platform as clean and minimal. However, minimalism sometimes came at the cost of clarity. Some screens were described as “too white” or lacking in visual hierarchy. Font styles were noted as too light in some places, making reading difficult. Others suggested subtle visual accents, clearer section titles, or more distinct color cues to help guide the eye. The overall design was functional, but could benefit from more visual depth.

Task Flows and Forms

Users completed tasks like creating a bounty, editing a post, and submitting agreements, but often stumbled on smaller interface elements. In several cases, key buttons were either too small or not clearly labeled. Tag entry fields caused confusion, some expected hashtags, others used commas or spaces with mixed results. The markdown editor was generally seen as useful, but participants wanted clearer guidance on formatting tools and better spacing between form fields.

Feedback and Error Handling

System feedback was inconsistent. Some error messages were helpful, especially when form fields were left empty. But others were vague or barely visible. One participant mentioned that notifications flashed too quickly to read. A recurring suggestion was to display feedback messages for longer, preferably in a more visible spot like the center or sidebar, rather than tucked away at the bottom.

Security Perception

Security was a recurring theme in the interviews. Most participants felt the inclusion of two-factor authentication added credibility. However, a few technical users noticed flaws like being able to change a password without proper validation or using the same login credentials across different roles. These loopholes made the platform feel less secure to them, despite the presence of other security features.

Participant Ratings

 P#	Overall Experience (1-10)	Security Perception (1-10)	Visual Design (1-10)	Navigation Ease (1-10)	Onboarding Experience (1-10)	Would you Recommend
P1	3.0	5.0	3	5	5	Maybe (if improved)
P2	4.0	4.5	4	6	6	Yes (after fixes)
P3	7.5	7.0	7	8	7	Yes
P4	9.0	9.0	8	8	9	Yes (after fixes)
P5	7.5	6.0	8.5	6.0	5.0	Yes (after fixes)
P6	9	9.0	8.0	9.0	9.0	Yes
Average Score (out of 10)	6.67	6.75	6.42	7.0	6.83	Apr 2, 2025

Action Items by Participants

P1

- Implement a "Forgot Password" feature that uses two-factor authentication.
- Add a preview option for the submission before posting.
- Improve the visual design and layout of the "Submissions" page, including adding user profile information.
- Enhance the password change functionality, including the ability to reset password with a one-time password (OTP).
- Rephrase and reorganize the navigation and section titles to improve the overall intuitiveness of the platform.

P2

- Improve the instructions and color contrast for the onboarding process.
- Make the checkbox for the user agreement more prominent.
- Implement a sidebar or expandable menu for the organization-level dashboard.
- Ensure that users cannot log in to the organization dashboard using their personal credentials.
- Improve the visual design and readability of the organization-level dashboard.
- Implement the ability to zoom and navigate within the submissions section.
- Ensure that password changes require a minimum level of complexity.

P3

- Create a more visually appealing and engaging design with the addition of color and animations.
- Implement a more user-friendly onboarding process with clear guidance for first-time users.
- Improve the chat functionality to make it less overwhelming.

P4

- Improve the instructions for the QR code scanning process.
- Consider adding a dashboard or menu to help users navigate the platform better.
- Modify the disclosure form to open as a separate screen instead of a popup.
- Add a visual cue or message to indicate an empty homepage.
- Implement a way to easily add and select tags/keywords for bounty postings.
- Extend the duration of the success feedback message when creating a bounty.
- Separate the archived bounty postings from the active ones in the submissions list.
- Implement a search feature for the submissions list.
- Explore adding more color variations and font styles to the UI.

Participant Insights

 P#	Common Pain Points	Suggested Improvements
P1	Unclear UI labels, missing confirmation messages, cluttered layout	Better labeling, password recovery, highlight submission areas
P2	Too much white space, login mismatch, weak feedback messaging	Sidebar navigation, clearer error prompts, refine security flow
P3	QR code confusion, slight dashboard clutter, limited color variety	Add visual dashboard menu, reposition submission elements
P4	QR code clarity, tag entry unclear, visual feedback too fast	Improve tag input hints, expand feedback time, UI polish

Overall Feedback

- Use clear, descriptive form labels supported by simple icons and contextual tooltips.
- Streamline navigation with sidebar or top menus and guide users through key flows like signup to dashboard and post creation to review.
- Add progress indicators or visual cues to help users understand where they are in a workflow.
- Improve error messaging by using plain, user-friendly language that clearly explains issues and how to fix them.
- Present two-factor authentication in a supportive, easy-to-understand way, with stronger guidance during setup.
- Make feedback messages more noticeable and allow them to remain visible for longer durations.
- Refine tag input functionality using chip-style input, autocomplete options, or clear instructions (e.g., “Press Enter to add”).
- Enhance the usability of the markdown editor with better spacing, layout, and clearer formatting options.
- Provide better visual hierarchy across pages, especially in dashboards, to reduce cognitive load.
- Introduce search functionality and hover-based descriptions to help users quickly locate and understand elements.
- Clearly distinguish between active and archived posts through visual styling or separation.
- Highlight user submissions, agreements, and actionable sections more effectively.
- Enable profile previews within submission views for context and clarity.

- Ensure responsive design across all devices with readable fonts and accessible prompts.
- Maintain consistent design language and UI behavior throughout the platform.
- Include an in-app feedback mechanism so users can easily share suggestions or report issues.

Next Steps: Integration of Feedback

The team will now review current findings to spot patterns and decide where design changes are needed, whether in user flows, security cues, or overall interface. These updates will be rolled into the system, followed by further usability sessions with a broader mix of participants. Then another testing round will be run to ensure the changes actually make a difference.